power supply

U1 POWER-JACK

12V

U2 7805(TO-220) — IN, OUT, GND

C3 100uF, C1 0.1u, C4 100uF, C2 0.1u

5V

GND

POWER LED

5V — R2 330ohm — LED4 — GND

microcontroller unit

KEY1 K4-6×6_TH — 5V — RESET

RESET — C9 10uF — 5V

RESET — R17 10K — GND

U3 AT89S52-24PU

1	T2/P1.0	VCC	40 — 5V
2	T2EX/P1.1	P0.0/AD0	39
3	P1.2	P0.1/AD1	38
4	P1.3	P0.2/AD2	37
5	P1.4	P0.3/AD3	36
6	MOSI/P1.5	P0.4/AD4	35
7	MISO/P1.6	P0.5/AD5	34
8	SCK/P1.7	P0.6/AD6	33
9	RST	P0.7/AD7	32
10	RXD/P3.0	#EA/VPP	31 — 5V
11	TXD/P3.1	ALE/#PROG	30 — ALE/PROG
12	INT0/P3.2	#PSEN	29 — PSEN
13	INT1/P3.3	P2.7/A15	28 — MTR1_EN
14	T0/P3.4	P2.6/A14	27 — MTR1_IN_1
15	T1/P3.5	P2.5/A13	26 — MTR1_IN_2
16	#WR/P3.6	P2.4/A12	25 — MTR2_IN_1
17	#RD/P3.7	P2.3/A11	24 — MTR2_IN_2
18	XTAL2	P2.2/A10	23 — MTR2_EN
19	XTAL1	P2.1/A9	22
20	GND	P2.0/A8	21

SSD_A, SSD_B, SSD_C, SSD_D

MOSI, MISO, SCK, RESET

RX, TX

DISPLAY_TEST

BLANKING_INPUT — Red, yellow, green

C7 30pF, C8 30pF, X1 16MHz — GND

GND

J1 / J9 / J10

J1: MOSI, MISO, SCK, ALE/PROG, PSEN, RESET, 5V, GND

J9: RX

J10: TX

MOTOR DRIVER

5V — C6 10uF, C5 0.1uF — GND

P1 CONN-TH_2P-P5.00

U4 L293D

1	ENABLE1	VSS	16 — 5V
2	INPUT1	INPUT4	15 — MTR2_IN_2
3	OUTPUT1	OUTPUT4	14
4	GND	GND	13
5	GND	GND	12 — GND
6	OUTPUT2	OUTPUT3	11
7	INPUT2	INPUT3	10 — MTR2_IN_1
8	VS	ENABLE2	9 — MTR2_EN

MTR1_EN, MTR1_IN_1, MTR1_IN_2, 12V

P2 CONN-TH_2P-P5.00

7-Segment Display

R16 330ohm, R7 330ohm, R3 330ohm, R4 330ohm, R5 330ohm, R6 330ohm, R8 330ohm

5V — C11 10uF, C10 0.1uF — GND

U5 CD4511BEE4

7	A	VSS	8 — GND
1	B	LESTROBE	5 — GND
2	C	LT	3
6	D	BL	4
		VDD	16 — 5V

SSD_A, SSD_B, SSD_C, SSD_D

DISPLAY_TEST, BLANKING_INPUT

R9 1k, R10 1k, R11 1k, R12 1k — GND

Trafic Light section

Red — R1 330ohm — LED1 — GND

yellow — R13 330ohm — LED2 — GND

green — R14 330ohm — LED3 — GND

SEG1

A, B, C, D, E, F, G, DP — G2, G1 — GND

TITLE: Sheet_1		REV: 1.0
EasyEDA	Company: custom project	Sheet: 1/1
	Date: 2026-10-09 Drawn By: stamilarasan	